

Regression Tree: Motivation

Previously, we used a **classification tree** to predict class labels.

Classification Tree Output

Class label: Yes/No, Spam/Not Spam, Pass/Fail

But sometimes the output is numerical.

Regression Tree Output

Continuous value: price, marks, temperature, drug effectiveness

Main Idea

A regression tree predicts a number instead of a class.

Example Problem

We want to predict **Drug Effectiveness** from dosage and gender.

Dosage	Gender	Drug Effectiveness
4	M	0
6	M	2
10	F	3
15	F	100
20	M	100
25	M	100
30	F	20
35	F	10
40	M	0

Target Variable

Drug Effectiveness is numerical, so this is a regression problem.

Regression Tree vs Classification Tree

Topic	Classification Tree	Regression Tree
Output	Class label	Numerical value
Example	Yes/No	Drug effectiveness
Leaf value	Majority class	Average target value
Split criterion	Gini / Entropy	SSR / MSE
Goal	Pure classes	Low prediction error

Key Difference

In a regression tree, each leaf predicts the **average** target value of training samples in that leaf.

How Does a Regression Tree Predict?

Suppose a leaf contains drug effectiveness values:

0, 2, 3

The prediction for that leaf is the average:

$$\hat{y} = \frac{0 + 2 + 3}{3} = 1.67$$

Important

Every sample reaching this leaf will get prediction:

$$\hat{y} = 1.67$$

So

Regression tree prediction = average value in the leaf.

How Do We Choose a Split?

For classification trees, we wanted pure class groups.

For regression trees, we want groups where the numerical values are close to each other.

Good Regression Split

A good split creates child nodes where target values are similar.

Example

0, 2, 3

is a good group because values are close.

0, 100, 3

is a bad group because values are far apart.

Residual and SSR

For each sample:

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

Regression trees measure error using **Sum of Squared Residuals**:

$$\text{SSR} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- ▶ y_i = actual value
- ▶ $\hat{y}_i$ = predicted value

Rule

Choose the split with the lowest SSR.

Testing a Categorical Split: Gender

Split the data using **Gender**.

Male Group

Effectiveness: 0, 2, 100, 100, 0

Female Group

Effectiveness: 3, 100, 20, 10

$$\hat{y}_M = \frac{0 + 2 + 100 + 100 + 0}{5} = 40.4 \quad \hat{y}_F = \frac{3 + 100 + 20 + 10}{4} = 33.25$$

$$\text{SSR}(\text{Gender}) = 17929.95$$

Testing a Numerical Split: Dosage

For numerical features, we test thresholds. Sorted dosages:

4, 6, 10, 15, 20, 25, 30, 35, 40

Candidate thresholds are midpoints:

5, 8, 12.5, 17.5, 22.5, 27.5, 32.5, 37.5

Example

Between dosage 20 and 25:

$$\frac{20 + 25}{2} = 22.5$$

So one possible split is Dosage < 22.5.

Example Split: Dosage < 22.5

Left Group (Dosage < 22.5)

Effectiveness: 0, 2, 3, 100, 100

$$\hat{y}_L = \frac{0 + 2 + 3 + 100 + 100}{5} = 41$$

Right Group (Dosage $\geq$ 22.5)

Effectiveness: 100, 20, 10, 0

$$\hat{y}_R = \frac{100 + 20 + 10 + 0}{4} = 32.5$$

$$\text{SSR}(\text{Dosage} < 22.5) = 17883$$

Compare All First-Level Splits

Split	SSR
Gender	17929.95
Dosage < 5	16485
Dosage < 8	14670
Dosage < 12.5	12355
Dosage < 17.5	17177
Dosage < 22.5	17883
Dosage < 27.5	14709
Dosage < 32.5	15374
Dosage < 37.5	16485

Best First Split

Lowest SSR is for:

Dosage < 12.5

First Split of the Regression Tree

Since **Dosage** < 12.5 has the lowest SSR, it becomes the root.



$$\hat{y}_{\text{left}} = \frac{0 + 2 + 3}{3} = 1.67$$

$$\hat{y}_{\text{right}} = \frac{100 + 100 + 100 + 20 + 10 + 0}{6}$$

$$\hat{y}_{\text{right}} = 55$$

Continue Splitting the Right Branch

Stopping Rule

We stop splitting the left branch because it has only 3 samples. In practice, regression trees use a minimum-samples rule to avoid overfitting.

The left branch has values 0, 2, 3 — these are already close. The left branch has values 0, 2, 3 and only 3 samples. Since the node is small and the values are close, we stop splitting it. The right branch has:

100, 100, 100, 20, 10, 0

This branch is still mixed, so we try to split it again.

Candidate Splits in Right Branch

Dosage < 17.5, 22.5, 27.5, 32.5, 37.5

Choosing the Second Split

For the right branch ($\text{Dosage} \geq 12.5$):

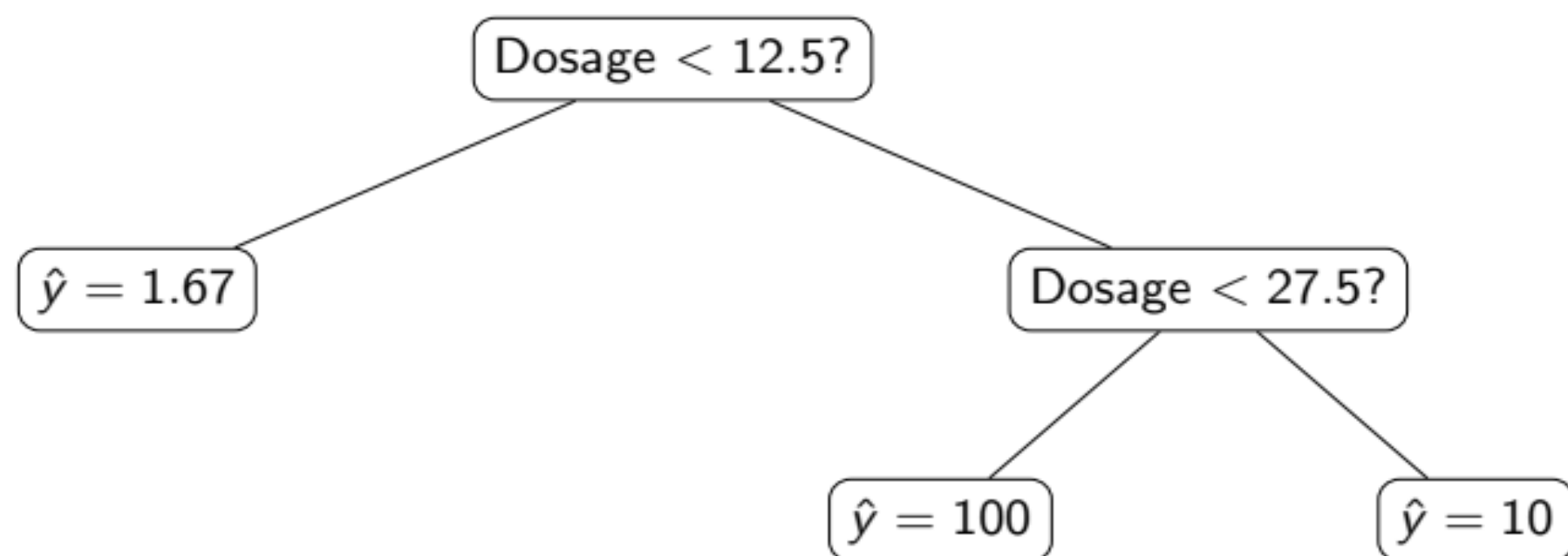
Candidate Split	SSR
Dosage < 17.5	9920
Dosage < 22.5	6275
Dosage < 27.5	200
Dosage < 32.5	7425
Dosage < 37.5	13076
Gender	10990.45

Best Second Split

Lowest SSR is:

Dosage < 27.5

Final Regression Tree



Leaf Predictions

$$\text{Dosage} < 12.5 \Rightarrow \hat{y} = 1.67$$

$$12.5 \leq \text{Dosage} < 27.5 \Rightarrow \hat{y} = 100$$

$$\text{Dosage} \geq 27.5 \Rightarrow \hat{y} = 10$$

Prediction Example

Suppose a new patient receives dosage = 20.

Use the tree:

1. Is Dosage < 12.5? No.
2. Is Dosage < 27.5? Yes.

Prediction

$$\hat{y} = 100$$

The predicted drug effectiveness is **100**.

Regression Tree Construction Algorithm

1. Start with all training data.
2. Try all possible splits.
3. For each split, compute leaf averages.
4. Calculate SSR for the split.
5. Choose the split with the lowest SSR.
6. Repeat for each child node.
7. Stop when the node is too small or no useful split remains.

Main Rule

Best Split = Split with minimum SSR

Summary

- ▶ Regression trees predict **numerical** values.
- ▶ Each leaf predicts the **average** target value.
- ▶ Error is measured using **residuals**.
- ▶ Splits are chosen by minimising **SSR** (or MSE).
- ▶ The best split has the **lowest SSR**.
- ▶ Numerical features use **thresholds**; categorical features use **categories**.

Regression Tree = Questions + Averages +
Minimum SSR

Example: Regression Tree with Two Important Features

Now consider a new dataset where both **Dose** and **Gender** affect the target.

Dose	Gender	Drug Effectiveness
5	M	10
6	M	12
7	F	20
8	F	22
20	M	80
21	M	82
22	F	95
23	F	97

Observation

Higher dose increases effectiveness. Also, for the same dose range, female patients have higher effectiveness than male patients.

Step 1: Try First-Level Splits

For numerical feature **Dose**, candidate thresholds are midpoints:

5.5, 6.5, 7.5, 14, 20.5, 21.5, 22.5

We also test the categorical split:

$$\textit{Gender} = M/F$$

Rule

For regression trees, choose the split with the lowest SSR.

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Step 2: Compare First-Level Splits

Candidate Split	SSR
Dose < 5.5	8805.43
Dose < 6.5	6308.00
Dose < 7.5	3822.80
Dose < 14	333.00
Dose < 20.5	3513.47
Dose < 21.5	5741.33
Dose < 22.5	8556.86
Gender	10533.00

Best Root Split

The lowest SSR is:

$$Dose < 14$$

So **Dose** becomes the root node.

Step 3: Tree After Root Split

Using the root split:

$$Dose < 14$$

we get two branches.

Left Branch: $Dose < 14$

Effectiveness values:

10, 12, 20, 22

Average prediction:

$$\hat{y} = \frac{10 + 12 + 20 + 22}{4} = 16$$

Right Branch: $Dose \geq 14$

Effectiveness values:

80, 82, 95, 97

Average prediction:

$$\hat{y} = \frac{80 + 82 + 95 + 97}{4} = 88.5$$

Important

Both branches are still mixed, so we try to split them again.

Step 4: Split the Left Branch

Left branch data:

Dose	Gender	Drug Effectiveness
5	M	10
6	M	12
7	F	20
8	F	22

Current prediction:

$$\hat{y} = 16$$

Current SSR:

$$\begin{aligned} & (10 - 16)^2 + (12 - 16)^2 + (20 - 16)^2 + (22 - 16)^2 \\ &= 36 + 16 + 16 + 36 = 104 \end{aligned}$$

Now we test whether splitting by **Gender** improves this branch.

Step 5: Left Branch Gender Split

Split the left branch by Gender.

Male Group

Effectiveness:

10, 12

Prediction:

$$\hat{y}_M = \frac{10 + 12}{2} = 11$$

SSR:

$$(10 - 11)^2 + (12 - 11)^2 = 2$$

Female Group

Effectiveness:

20, 22

Prediction:

$$\hat{y}_F = \frac{20 + 22}{2} = 21$$

SSR:

$$(20 - 21)^2 + (22 - 21)^2 = 2$$

$$SSR(\text{Gender}) = 2 + 2 = 4$$

Decision

SSR decreases from 104 to 4, so **Gender is useful** in the left

Step 6: Split the Right Branch

Right branch data:

Dose	Gender	Drug Effectiveness
20	M	80
21	M	82
22	F	95
23	F	97

Current prediction:

$$\hat{y} = 88.5$$

Current SSR:

$$\begin{aligned} & (80 - 88.5)^2 + (82 - 88.5)^2 + (95 - 88.5)^2 + (97 - 88.5)^2 \\ &= 72.25 + 42.25 + 42.25 + 72.25 = 229 \end{aligned}$$

Now we test whether **Gender** improves this branch.

Step 7: Right Branch Gender Split

Split the right branch by Gender.

Male Group

Effectiveness:

80, 82

Prediction:

$$\hat{y}_M = \frac{80 + 82}{2} = 81$$

SSR:

$$(80 - 81)^2 + (82 - 81)^2 = 2$$

Female Group

Effectiveness:

95, 97

Prediction:

$$\hat{y}_F = \frac{95 + 97}{2} = 96$$

SSR:

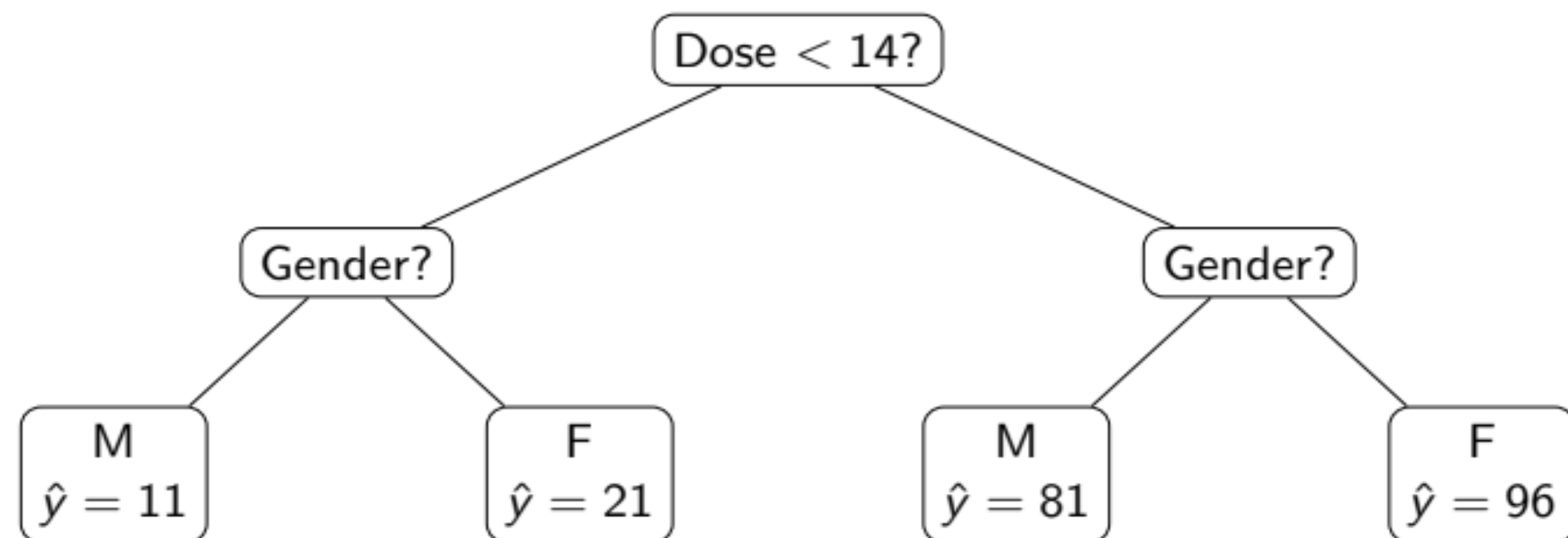
$$(95 - 96)^2 + (97 - 96)^2 = 2$$

$$SSR(\text{Gender}) = 2 + 2 = 4$$

Decision

SSR decreases from 229 to 4, so **Gender is also useful** in the

Final Regression Tree Using Two Features



Interpretation

First, the tree separates low-dose and high-dose patients. Then, inside each dose group, it uses Gender to make a more accurate prediction.

Final Rules

The regression tree can be written as rules:

Condition 1	Condition 2	Prediction
Dose < 14	Gender = M	$\hat{y} = 11$
Dose < 14	Gender = F	$\hat{y} = 21$
Dose $\geq$ 14	Gender = M	$\hat{y} = 81$
Dose $\geq$ 14	Gender = F	$\hat{y} = 96$

Interpretation

The final prediction is the average target value of the training samples in that leaf.

Prediction Example

Suppose a new patient has:

$$Dose = 22, \quad Gender = F$$

Use the tree:

1. Is $Dose < 14$? No.
2. $Gender = F$.

Prediction

$$\hat{y} = 96$$

The predicted drug effectiveness is 96.

Why This Example Shows Two Feature Effects

In the previous example, only Dosage appeared in the final tree. Here, both features appear:

$$\text{Dose} \rightarrow \text{Gender}$$

Meaning

Dose is useful because it separates low-effectiveness and high-effectiveness groups.

Meaning

Gender is useful because within each dose range, it further reduces SSR.

Important Point

A feature appears in the regression tree only if it reduces prediction error enough compared with other possible splits.